

MINIMUM DEGREE EIGHT IN K_7^- -MINOR-FREE CONTRACTION-CRITICAL GRAPHS

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ABSTRACT. Let G be a minor-minimal non-six-colourable K_7^- -minor-free graph, with n vertices and m edges. We prove that $n_7 = 0$, $\delta(G) \geq 8$, $m \geq 4n$, and G has no K_5 subgraph. If n_i counts the vertices of degree i and $\tau = \sum_{i \geq 10} (i - 9)n_i$, then $n_8 \geq 25 + \tau$; in particular, G has at least twenty-five degree-eight vertices, each with a K_4 -free neighbourhood. The capstone is a rooted-minor closure: a six-connected n -vertex graph with a degree- d vertex whose neighbourhood contains K_4 and with at least $4n + d - 13$ edges contains a K_7^- minor. We also give a direct uniform proof that, for every $r \geq 2$, an $(r + 1)$ -connected graph containing two distinct K_r subgraphs has a K_{r+2}^- minor. All proofs are computation-free.

1. INTRODUCTION

Hadwiger's conjecture states that every graph of chromatic number t contains a K_t minor [3]. The conjecture is open for $t = 7$. Write K_t^- for the graph obtained from K_t by deleting one edge, and write $K_7^\vee$ for the graph obtained from K_7 by deleting two incident edges. Every K_7^- -minor-free graph is seven-colourable [1, Theorem 1]; Albar's proof uses Jakobsen's extremal theorem. Norin and Totschnig proved that every $K_7^\vee$ -minor-free graph is six-colourable [8, Theorem 4]. They asked whether the same conclusion holds under the weaker exclusion of K_7^- [8, Conjecture 21].

Neighbourhoods of low-degree vertices in contraction-critical graphs have a long history. Mayer obtained a finite catalogue of candidate neighbourhood graphs for degree-seven vertices in six-contraction-critical graphs [7]. In the minor-minimal $K_7^\vee$ -minor-free setting, Norin and Totschnig proved that every degree-seven vertex belongs to a K_5 contained in its closed neighbourhood [8, Claim 4.4]. Here a rooted-minor argument first determines the entire neighbourhood. A second rooted argument, using their $K_{4,2}^*$ bound, then shows that no degree-seven vertex can occur at all.

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We first state the unconditional linkage theorem used in that reduction. Rolek and Song proved its case $r = 6$ [9, Lemma 1.9]; the argument below gives a direct uniform proof.

Theorem 1.1. *For every $r \geq 2$, every $(r + 1)$ -connected graph containing two distinct K_r subgraphs contains a K_{r+2}^- minor.*

For $2 \leq r \leq 6$, Theorem 1.1 also follows from Rolek and Song's result. Ad-join a universal clique of order $6 - r$. The resulting graph is seven-connected and contains two distinct K_6 subgraphs: deleting at most six vertices either leaves an added universal vertex or removes the whole added clique and at most r vertices of the original graph. Their lemma therefore gives a K_8^- model. Since the model bags are disjoint, at most $6 - r$ of them meet the added clique. Discard those bags and retain any $r + 2$ of the remaining bags; they form a K_{r+2}^- model. We include the direct proof because it applies for every $r \geq 2$ and gives the localisation in Lemma 2.1. The connectivity hypothesis is sharp: K_{r+1} is r -connected and contains two distinct K_r subgraphs, but has no K_{r+2}^- minor.

The following rooted closure is the main local tool. A graph is *minor-minimal non-six-colourable* if it is not six-colourable and every proper minor is six-colourable.

Theorem 1.2. *Let G be a six-connected graph of order n , let $v \in V(G)$ have degree $d \geq 5$, and suppose that $G[N(v)]$ contains a K_4 subgraph. If*

$$|E(G)| \geq 4n + d - 13,$$

then $K_7^- \preccurlyeq G$.

Our principal result gives the resulting critical-host structure.

Theorem 1.3. *Let G be a minor-minimal non-six-colourable K_7^- -minor-free graph. For each i , let n_i denote the number of vertices of degree i , and put*

$$\tau = \sum_{i \geq 10} (i - 9)n_i.$$

Then G is seven-connected and

$$n_7 = 0, \quad \delta(G) \geq 8, \quad |E(G)| \geq 4|V(G)|.$$

Moreover, G contains no K_5 subgraph and

$$n_8 \geq 25 + \tau.$$

Every degree-eight vertex of G has a K_4 -free neighbourhood.

Albar proved the analogous lower bound of twenty-five degree-eight vertices for a hypothetical minor-minimal non-seven-colourable K_7^- -minor-free graph [1, Lemma 6]. Theorem 1.3 concerns criticality one level lower: the rooted closure eliminates degree seven and gives the defect-sensitive bound $n_8 \geq 25 + \tau$.

A *minor model* of a graph H in G is a family $(B_x : x \in V(H))$ of pairwise disjoint connected vertex sets such that B_x and B_y are adjacent whenever $xy \in E(H)$. We use $H \preceq G$ to denote that H is a minor of G . All graphs are finite and simple.

2. TWO LINKED CLIQUES

We begin with the disjoint case.

Lemma 2.1. *Let $r \geq 2$. If an $(r+1)$ -connected graph H contains two vertex-disjoint K_r subgraphs, then $K_{r+2}^- \preceq H$. The branch sets can be chosen to meet the union of the two cliques.*

Proof. Write the two cliques as

$$A = \{a_1, \dots, a_r\}, \quad B = \{b_1, \dots, b_r\}.$$

By Menger's theorem there are r pairwise vertex-disjoint A to B paths whose open interiors avoid $A \cup B$. Relabel them as paths P_i from a_i to b_i , and put $U = \bigcup_i V(P_i)$.

Call a path Q a *usable cross-path* if its ends lie on distinct paths P_i, P_j , its open interior avoids U , and its ends are not both A -ends or both B -ends. Suppose first that Q exists. Split each of P_i and P_j across an edge into a nonempty A -prefix and a nonempty B -suffix, choosing the split so that the ends of Q belong to pieces of opposite kinds. Absorb the open interior of Q into the piece containing one end. The four resulting pieces are connected and disjoint. The two A -pieces are adjacent through A , the two B -pieces are adjacent through B , the two pieces from each original path are adjacent across its splitting edge, and Q supplies one of the two remaining cross-adjacencies. Thus at most one adjacency among the four pieces is missing. Each unsplit path is adjacent to every split piece through its two clique ends, and the unsplit paths are pairwise adjacent. These $r+2$ sets form a K_{r+2}^- model.

Suppose now that no usable cross-path exists. Let C be a component of $H - U$. If C has neighbours on two distinct paths, a path through C between two such attachments is unusable only when both attachments are A -ends or both are B -ends. Comparing all pairs of attachments shows that all neighbours of C are A -ends or all are B -ends. Hence $|N_H(C)| \leq r$, contrary to $(r+1)$ -connectivity. Every component of $H - U$ therefore attaches to only one P_i .

For each i , let W_i be the union of the open interior of P_i and all components of $H - U$ that attach to P_i . No vertex of W_i has a neighbour on P_j for $j \neq i$. Indeed, an edge from the open interior of P_i to P_j would be a usable cross-path, while such an edge from a component of $H - U$ would make that component attach to two distinct linkage paths, contrary to the preceding paragraph. Hence

$$N_H(W_i) \subseteq \{a_i, b_i\}.$$

Since $r \geq 2$, a nonempty W_i would give a vertex cut of order at most two. Thus every W_i is empty. It follows that H consists of A , B , and the matching edges $a_i b_i$: any other edge between A and B would be usable. This graph has minimum degree r , again contradicting $(r + 1)$ -connectivity. $\square$

Proof of Theorem 1.1. Let L_1, L_2 be distinct K_r subgraphs and put $s = |V(L_1) \cap V(L_2)|$. If $s \leq r - 2$, delete the common set Z . The graph $G - Z$ is $(r + 1 - s)$ -connected and contains the two disjoint K_{r-s} subgraphs $L_1 - Z$ and $L_2 - Z$. Lemma 2.1 gives a K_{r-s+2}^- model in which every branch set contains a vertex of $(L_1 \cup L_2) - Z$. The vertices of Z , taken as singleton branch sets, are complete to this model and to one another. They complete a K_{r+2}^- model.

It remains that $s = r - 1$. The union $X = V(L_1) \cup V(L_2)$ induces a graph containing K_{r+1}^- . Since an $(r + 1)$ -connected graph has at least $r + 2$ vertices, $G - X$ is nonempty. Let C be a component of $G - X$. If C missed a vertex of X , then $N_G(C)$ would be a cut of order at most r . Thus C is adjacent to every vertex of X . The vertices of X as singleton branch sets, together with C , form a K_{r+2}^- model. $\square$

The case $r = 5$ gives the following form used throughout the paper.

Corollary 2.2. *Every six-connected K_7^- -minor-free graph contains at most one K_5 subgraph.*

3. DEGREE-SEVEN NEIGHBOURHOODS

We use two standard facts about contraction-critical graphs. A noncomplete k -contraction-critical graph is seven-connected when $k \geq 7$ [6, Satz 6]; see also [9, Theorem 1.8]. Also, if G is k -contraction-critical and $v \in V(G)$, then

$$(3.1) \quad \alpha(G[N(v)]) \leq d(v) - k + 2;$$

This inequality goes back to Dirac [2]; we use the exact formulation in [9, Lemma 1.6(i)].

The following is an immediate specialization of Theorem 7 and the definition of property $(*)$ in Kriesell and Mohr [5, Theorem 7]. Let a graph J be properly coloured with five colours, and choose one root from each colour class. Let D be a graph on the five roots with at most six edges. If the ends of every edge $xy \in E(D)$ lie in the same bichromatic component of J on their two colours, then J contains pairwise disjoint connected sets, one containing each root, that are adjacent for every edge of D .

Lemma 3.1. *Let G be a seven-chromatic graph such that every proper minor is six-colourable and $K_7^- \not\leq G$. Let v have degree seven, put $H = G[N(v)]$, and let ab be a nonedge of H . If*

$$R = N(v) \setminus \{a, b\},$$

then $G - v - \{a, b\}$ contains a K_5 minor rooted at R .

Proof. Contract the two edges va and vb to one vertex. A six-colouring of the resulting proper minor induces a six-colouring φ of $G - v$ in which a and b have one common colour, say 0. Every vertex of R avoids colour 0. The five vertices of R use the other five colours distinctly, since a colour absent from $N(v)$ could be assigned to v .

Put $F = \overline{H}$. Inequality (3.1) gives $\alpha(H) \leq 2$, so F is triangle-free. If $xy \in E(F[R])$ and x, y have colours i, j , then x, y lie in the same component of the subgraph of $G - v$ induced by colours i, j . Otherwise, interchanging i and j on the component containing x would remove colour i from $N(v)$, allowing that colour at v .

Delete the colour-0 class from $G - v$. The resulting graph is properly five-coloured, with R a transversal of its colour classes. By Mantel's theorem, the triangle-free graph $F[R]$ has at most six edges. The Kriesell–Mohr result, applied with demand graph $F[R]$, gives five disjoint connected sets rooted at R and adjacent for every nonedge of $H[R]$. Every other pair is joined by its edge in $H[R]$, so the five sets form the required rooted K_5 model. $\square$

Theorem 3.2. *Under the hypotheses of Lemma 3.1, every degree-seven vertex v satisfies*

$$(3.2) \quad G[N(v)] \cong K_4 \dot{\cup} K_3 \quad \text{or} \quad K_1 \vee (K_3 \dot{\cup} K_3).$$

In particular, v belongs to a K_5 subgraph.

Proof. Put $H = G[N(v)]$ and $F = \overline{H}$. By (3.1), F is triangle-free. It is nonempty, since otherwise $G[N(v)]$ would contain K_7^- .

Every nonisolated vertex of F has degree at least three. To see this, choose an edge $ab \in E(F)$ and use the rooted model of Lemma 3.1 on $R = N(v) \setminus \{a, b\}$. If $d_F(a) = 1$, then a is adjacent in H to every root; the rooted model, $\{a\}$, and $\{v\}$ form a K_7 model. If $d_F(a) = 2$, with other neighbour x , then triangle-freeness gives $bx \in E(H)$. Absorb b into the branch set rooted at x . Together with $\{a\}$ and $\{v\}$, the resulting seven branch sets have at most one missing adjacency, again a contradiction.

Every nontrivial component of F therefore has minimum degree at least three and hence at least six vertices. Since $|V(F)| = 7$, there is one nontrivial component and at most one isolated vertex. If the component has six vertices, Mantel's theorem and its equality case give $F \cong K_{3,3} \dot{\cup} K_1$.

Suppose instead that the component has seven vertices. No vertex has degree at least five: its independent neighbourhood would contain a vertex of degree at most two. Thus every degree is three or four. The degree sum is even, so some vertex z has degree four. Its four neighbours are independent. Each is adjacent to both vertices outside $N_F[z]$ in order to have degree at least three, and those two outside vertices are nonadjacent by triangle-freeness. Hence $F \cong K_{3,4}$. Taking complements gives (3.2). $\square$

For a minor-minimal non-six-colourable K_7^- -minor-free graph, Albar's theorem gives $\chi(G) = 7$, while Mader's theorem gives seven-connectivity. Corollary 2.2, together with Theorem 3.2, gives the sharper conclusion below.

Corollary 3.3. *Let G be as in Theorem 1.3. Then G contains at most one K_5 subgraph and every degree-seven vertex v satisfies*

$$G[N(v)] \cong K_4 \dot{\cup} K_3.$$

Proof. The only point not already stated is the exclusion of the second graph in (3.2). In that graph, v belongs to two distinct K_5 subgraphs, which contradicts Corollary 2.2. $\square$

4. ROOTED TWO-HELPER CLOSURE

For a graph F and $S \subseteq V(F)$, say that (F, S) is *internally k -connected* if there is no separation (A, B) of F with $S \subseteq A$, $B \setminus A \neq \emptyset$, and $|A \cap B| < k$. A Z -rooted $K_{4,2}^*$ model consists of four pairwise disjoint connected root bags, one containing each member of the four-set Z , and two further disjoint connected helper bags. Each helper is adjacent to all four root bags, and the helpers are adjacent to one another; no adjacency between distinct root bags is required.

We use the following rooted bound of Norin and Totschnig [8, Lemma 12].

Theorem 4.1 (Norin–Totschnig). *Let F be a graph and let $Z \subseteq V(F)$ have order four. If (F, Z) is internally four-connected and F has no Z -rooted $K_{4,2}^*$ model, then*

$$(4.1) \quad |E(F)| \leq 4|V(F)| - 10.$$

The next elementary augmentation places one additional prescribed vertex in a helper bag.

Lemma 4.2. *Let F be a graph and let $S = Z \dot{\cup} \{x\}$, where $|Z| = 4$. Suppose that (F, S) is internally five-connected. If F has a Z -rooted $K_{4,2}^*$ model, then it has one in which x belongs to a helper bag.*

Proof. Choose a Z -rooted model whose helper bags U, V have maximum total order and, subject to that, whose root bags $R_1, \dots, R_4$ have minimum total order. Write z_i for the root in R_i , and put

$$P_i = \{r \in R_i : r \text{ has a neighbour in } U \cup V\}.$$

We claim that $|P_i| = 1$. Both helpers meet R_i , so P_i is nonempty. If it had at least two vertices, there would be distinct $u, w \in R_i$ such that u meets U and w meets V ; otherwise both helper-contact sets would be the same singleton. Take a minimal tree in $F[R_i]$ containing z_i, u, w . By the minimal choice of the root bags, this tree spans R_i . One of u, w , say u , is a leaf different from z_i . Moving u from R_i into U preserves connectivity of both altered bags. The former tree edge from u to $R_i \setminus \{u\}$ preserves the R_i – U contact, while w preserves the R_i – V contact. Every other required adjacency is unchanged, contradicting maximality of the helper union. Hence $|P_i| = 1$.

No component outside the six model bags can meet $U \cup V$, since it could be absorbed into a helper that it meets. It follows that

$$(4.2) \quad |N_F(U \cup V) \setminus (U \cup V)| \leq 4,$$

with at most one external neighbour in each root bag. If x were outside the helper bags, (4.2) would give a separation of (F, S) of order at most four whose second open side is the nonempty set $U \cup V$. This contradicts internal five-connectivity. Thus $x \in U \cup V$. $\square$

Proof of Theorem 1.2. Choose a four-set $Z \subseteq N_G(v)$ inducing K_4 , choose $x \in N_G(v) \setminus Z$, and put $F = G - v$. Deleting one vertex from a six-connected graph leaves a five-connected graph. In particular, (F, Z) is internally four-connected and $(F, Z \cup \{x\})$ is internally five-connected. Moreover,

$$(4.3) \quad \begin{aligned} |E(F)| &= |E(G)| - d \\ &\geq 4n - 13 = 4|V(F)| - 9. \end{aligned}$$

Theorem 4.1 supplies a Z -rooted $K_{4,2}^*$ model in F , and Lemma 4.2 lets us choose it with x in one helper bag.

The four root bags are pairwise adjacent through the literal edges of $G[Z] = K_4$. Together with the two adjacent helpers they form a K_6 model. The singleton bag $\{v\}$ is adjacent to all four root bags and to the helper containing x ; it may miss only the other helper. These seven bags form a K_7^- model in G . $\square$

5. CRITICAL-HOST CONSEQUENCES

We first record the density input. Jakobsen proved that an n -vertex graph with $n \geq 7$ and at least $\frac{9}{2}n - 12$ edges contains a K_7^- minor or is a $(K_{2,2,2,2}, K_6, 4)$ -cockade [4, Theorem 4]; see also [1, Theorem 2].

Lemma 5.1. *If G is as in Theorem 1.3, with $n = |V(G)|$ and $m = |E(G)|$, then*

$$(5.1) \quad 2m \leq 9n - 25.$$

Proof. The graph G is not either base graph of the cockade family, since both K_6 and $K_{2,2,2,2}$ are six-colourable. A nontrivial cockade has a vertex cut of order four, whereas G is seven-connected. Thus G is not an exceptional cockade. Since it has no K_7^- minor, Jakobsen's theorem gives $m < \frac{9}{2}n - 12$. Integrality yields (5.1). $\square$

The first density bound uses only the linked-clique and neighbourhood results above.

Proposition 5.2. *Let G be as in Theorem 1.3. Then*

$$(5.2) \quad n_7 \leq 5, \quad |E(G)| \geq 4|V(G)| - 2.$$

Proof. By Corollary 3.3, every degree-seven vertex lies in the unique possible K_5 , so $n_7 \leq 5$. Seven-connectivity gives minimum degree at least seven, and every vertex not counted by n_7 has degree at least eight. Hence

$$2|E(G)| \geq 7n_7 + 8(|V(G)| - n_7) = 8|V(G)| - n_7 \geq 8|V(G)| - 5.$$

The left side is even, so it is at least $8|V(G)| - 4$, proving (5.2). $\square$

Proof of Theorem 1.3. Albar's theorem gives $\chi(G) = 7$, and every proper minor is six-colourable by the choice of G . The graph is noncomplete because it has no K_7^- minor, so Mader's theorem gives seven-connectivity.

Suppose that v has degree seven. Corollary 3.3 shows that $G[N(v)]$ contains a K_4 . Proposition 5.2 gives

$$|E(G)| \geq 4|V(G)| - 2 > 4|V(G)| - 6,$$

which is more than the threshold in Theorem 1.2 when $d_G(v) = 7$. That theorem gives the forbidden K_7^- minor. Thus $n_7 = 0$. Seven-connectivity now gives $\delta(G) \geq 8$, and degree summation gives $|E(G)| \geq 4|V(G)|$.

Put $q = |E(G)| - 4|V(G)|$, so $q \geq 0$. Suppose that a K_5 subgraph has vertex set K . For each $w \in K$, the neighbourhood of w contains the K_4 on $K \setminus \{w\}$. Since G has no K_7^- minor, the contrapositive of Theorem 1.2 gives

$$4|V(G)| + q < 4|V(G)| + d_G(w) - 13,$$

and hence $d_G(w) \geq q + 14$. The five vertices of K would contribute at least $5(q + 6)$ to

$$\sum_{z \in V(G)} (d_G(z) - 8) = 2q,$$

which is impossible. Thus G contains no K_5 subgraph.

Finally, Lemma 5.1 and the exact degree identity give

$$25 \leq 9|V(G)| - 2|E(G)| = n_8 - \tau.$$

Therefore $n_8 \geq 25 + \tau$. If a degree-eight vertex had a K_4 in its neighbourhood, it would belong to a K_5 , which has just been excluded. $\square$

Theorem 1.3 reduces the Norin–Totschnig conjecture to an extremal statement only in the $4n$ range.

Corollary 5.3 (Extremal reduction). *Suppose that every seven-connected graph H satisfying*

$$|E(H)| \geq 4|V(H)|$$

contains a K_7^- minor. Then every K_7^- -minor-free graph is six-colourable.

Proof. Otherwise, choose a minor-minimal non-six-colourable K_7^- -minor-free graph G . By Theorem 1.3, the graph G is seven-connected and $|E(G)| \geq 4|V(G)|$. The assumed extremal statement therefore gives $K_7^- \preceq G$, a contradiction. $\square$

The extremal statement in Corollary 5.3 remains open. Thus the present paper does not prove the Norin–Totschnig conjecture, and it does not settle Hadwiger's conjecture for chromatic number seven.

STATEMENT ON AI USE AND INDEPENDENCE

This research was conducted independently by Cavit Erginsoy using GPT-5.6 Sol, a model developed by OpenAI. GPT-5.6 Sol contributed substantially to theorem discovery, proof development, verification, and manuscript preparation. Erginsoy directed and curated the research programme. Erginsoy has no affiliation with OpenAI; OpenAI did not sponsor, supervise, endorse, or verify this work.

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